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Automated Internal Quality Audit Using a Cross-Modal RAG Agent: A Benchmark of Qwen, DeepSeek-R1, and Mistral on 19 LAM-Teknik Indicators

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ABSTRACT (IN ENGLISH)

Internal Quality Audit (AMI) is critical for Indonesian higher-education accreditation, yet study programs still score 60+ indicators manually by cross-referencing a 200-page Self-Evaluation Report (LED) and a 50-sheet Study Program Performance Report (LKPS), a process that takes 3–5 days per program and varies between assessors. This study develops an automated quality-assessment agent that ingests both documents and benchmarks three open-weight Large Language Models — Qwen 3.5 35B, DeepSeek-R1 32B, and Mistral-Small 3.2 24B — on 19 LAM-Teknik indicators sampled across all nine accreditation criteria (9 qualitative, 6 quantitative, 4 composite). The reference scores are produced by Claude as an LLM-as-annotator and validated against post-audit human assessments (89.5% exact agreement). The results show that Qwen 3.5 attains the lowest MAE (0.605) and RMSE (0.769) with 100% within-±1 accuracy and 32.5 s per indicator; DeepSeek-R1 is the most cautious but the slowest (MAE 1.395; 79.9 s) and collapses on quantitative items (MAE 2.17); Mistral is the fastest (13.6 s) and strong on quantitative items (MAE 0.50) yet exhibits a 78% false-refusal rate on qualitative ones. The agent reduces AMI cycle time from days to under one hour while preserving an auditable evidence trace.

Keywords: *Internal Quality Audit; LAM Teknik accreditation; cross-modal RAG; large language models; benchmarking*

ABSTRAK (DALAM BAHASA INDONESIA)

Audit Mutu Internal (AMI) merupakan kunci keberhasilan akreditasi pendidikan tinggi di Indonesia, namun program studi masih menilai 60+ indikator secara manual dengan mencocokkan LED 200 halaman dan LKPS 50 *sheet*, suatu proses yang menghabiskan 3–5 hari per program dan bervariasi antar asesor. Penelitian ini mengembangkan agen penilaian mutu otomatis yang membaca kedua dokumen sekaligus dan melakukan *benchmark* tiga *Large Language Model open-weight* — Qwen 3.5 35B, DeepSeek-R1 32B, dan Mistral-Small 3.2 24B — pada 19 indikator LAM Teknik yang mencakup sembilan kriteria akreditasi (9 kualitatif, 6 kuantitatif, 4 komposit). Skor referensi dihasilkan oleh Claude sebagai *LLM-as-annotator* dan divalidasi terhadap hasil asesmen pasca-audit manusia (kesepakatan *exact* 89,5%). Hasil menunjukkan Qwen 3.5 unggul dengan MAE terendah (0,605) dan RMSE (0,769), akurasi *within*-±1 sebesar 100%, serta 32,5 detik per indikator; DeepSeek-R1 paling konservatif tetapi paling lambat (MAE 1,395; 79,9 detik) dan kolaps pada indikator kuantitatif (MAE 2,17); Mistral tercepat (13,6 detik) dan kuat pada indikator kuantitatif (MAE 0,50), tetapi mengalami *false refusal* 78% pada indikator kualitatif. Agen ini memangkas waktu AMI dari beberapa hari menjadi kurang dari satu jam dengan jejak bukti yang dapat ditelusuri.

Kata kunci: *Audit Mutu Internal; akreditasi LAM Teknik; RAG lintas-modal; large language models; benchmarking*

INTRODUCTION

Higher-education accreditation in Indonesia operates under increasingly stringent quality-standardization demands, in which the Internal Quality Assurance System (SPMI) — through its operational arm, the Internal Quality Audit (AMI) — must ensure that every study program meets the National Standards for Higher Education (SN-Dikti) and the criteria set by BAN-PT or LAM (Efendi, Hapsari, & Yusup, 2025; Sakuroh, Marlina, Krismayanti, & Adiningsih, 2026; Aljarallah & Dutta, 2022). The primary obstacle is the reliance on manual assessment that produces fragmented "data silos" and delayed reporting (Efendi et al., 2025), particularly for technical study programs that require rigorous and continuous quality monitoring (Subijanto et al., 2021; Sugiarti, 2022); earlier research therefore explored data-driven alternatives such as K-Means clustering and OLAP for accreditation analytics (Sakinah, Syaputra, Rahman, Fajri, & Rachmansyah, 2025), AI-assisted self-evaluation frameworks (Ticolau, 2026), and data-lakehouse architectures for academic metrics (Isnaeni, Putranto, Andriyani, & Khomsah, 2025).

Recent studies converge on the conclusion that conventional internal-audit processes cannot capture the dynamics of large and fast-paced academic data

(Isnaeni et al., 2025); a comprehensive automated architecture that directly processes academic metrics is needed (Ticolau, 2026; Başaran, 2026), and the integration of automated data-analysis technology is expected to minimize human error and improve institutional transparency (Sakuroh et al., 2026). Large Language Models (LLMs) open new opportunities for automated processing of complex academic data (Naveed, Khan, & Qui, 2025; Demir & Yavuz, 2026), but model behavior diverges sharply across families: **Qwen 3.5** is an Alibaba dense decoder with strong multilingual instruction-following and native tool-use through `bind_tools` (Bai et al., 2023); it demonstrates high precision in error minimization (Aydin, Karaaslan, Erenay, & Bacanin, 2026). **DeepSeek-R1** is a reasoning-distilled model that emits an explicit `<think>...</think>` chain before answering and therefore cannot use native function calling on Ollama, requiring prompt-based JSON dispatch instead (DeepSeek-AI, 2025; Jegham, Abdelatti, & Hendawi, 2025). **Mistral-Small 3.2** is a 24 B mixture-of-experts variant emphasizing throughput; it offers the lowest latency among the three but is prone to hallucination on Indonesian-language tasks (Mistral AI, 2024; Nurohim, Setyadi, & Fauzi, 2025).

Research Questions

To address the gap above, this study answers three research questions:

- **RQ1.** Can a cross-modal RAG agent (LED PDF + LKPS Excel) reduce AMI cycle time for LAM-Teknik study programs while preserving an auditable evidence trace?
- **RQ2.** Among Qwen 3.5, DeepSeek-R1, and Mistral-Small, which model offers the best trade-off between accuracy, calibration, and runtime on a multi-criteria indicator sample?
- **RQ3.** How does scoring performance vary across indicator types (qualitative, quantitative, composite), and what does the variation imply for prompt and tool design?

Research Objectives and Novelty

This research aims to (1) design a cross-modal RAG agent that synchronously cross-references qualitative LED PDF evidence with quantitative LKPS Excel data through four specialized tools; (2) benchmark three open-weight LLMs against Claude-Sonnet reference scores on 19 indicators spanning all nine LAM-Teknik criteria; and (3) characterize per-type error patterns, calibration, and runtime to derive deployment guidance for Indonesian quality-assurance offices. The novelty of this study lies in four complementary contributions not addressed by the prior LLM-assessment literature, which has largely focused on English-language student-work grading (Sun et al., 2025) and single-modality RAG-in-education surveys (Gao et al., 2025): a *domain* contribution as the first open-weight LLM-agent study on Indonesian LAM-style accreditation; a *cross-modal RAG* contribution that joins

narrative PDF and structured Excel evidence in a single agent state; a *tool-calling paradigm* contribution that contrasts native function calling (Qwen, Mistral) with prompt-based JSON dispatch (DeepSeek-R1) on the same task; and a *calibration* contribution that documents confidence-error inversion in DeepSeek-R1 and false-refusal-with-overconfidence in Mistral on Indonesian institutional documents, extending the calibration literature (Lin, Hilton, & Evans, 2022; Geng, Yang, Cao, Zhang, & Tang, 2024) to a new domain.

METHODS

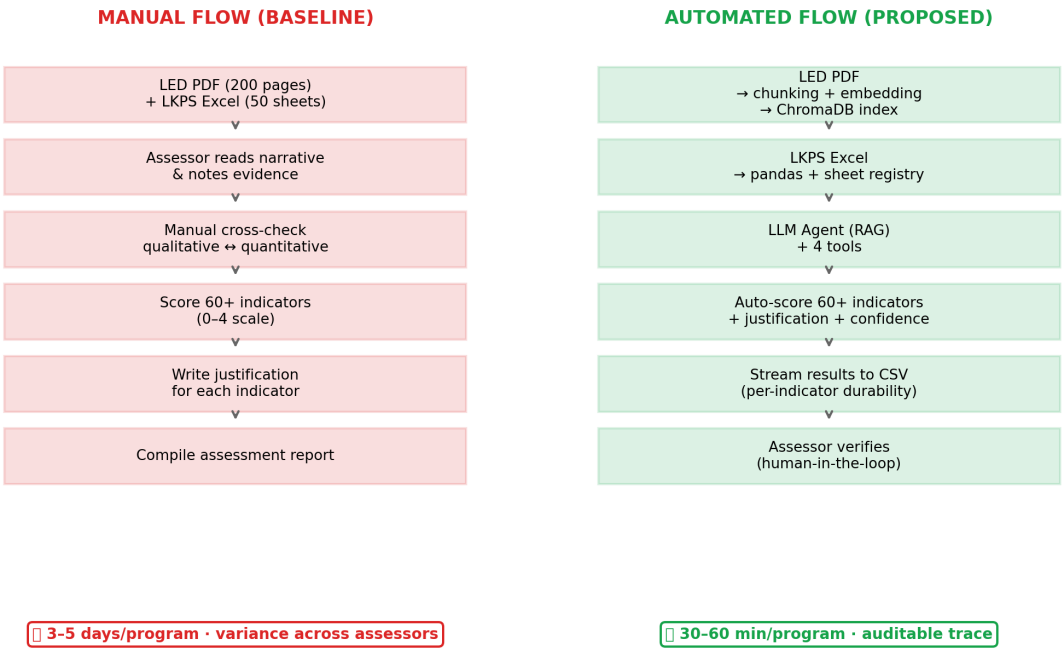
Approach and Scope. This study uses a quantitative experimental approach with single-run benchmarking against three open-weight LLMs. To address the limitation of an earlier Tri-Dharma-only sample, the scope is broadened to 19 indicators (≈32% of the 60-indicator LAM-Teknik instrument), purposively stratified across all nine criteria (VMTS, Tata Pamong, Mahasiswa, SDM, Keuangan-Sarpras, Pendidikan, Penelitian, PkM, Luaran) and across all three score-type families: 9 qualitative, 6 quantitative, and 4 composite indicators. The minimum cell size of n ≥ 4 per type permits a stable per-type breakdown.

Operational Variable Definitions. The primary dependent variable is the model score (ordinal 0–4) for each indicator, compared against the Claude reference score. Supporting variables are the justification text, the model-stated confidence (0–1), the runtime per indicator (seconds), and the indicator type (qualitative / quantitative / composite). Independent variables are the model (Qwen 3.5 35B / DeepSeek-R1 32B / Mistral-Small 3.2 24B) and the indicator type.

Location and Sample. The study was conducted on one Applied Bachelor in Telecommunication Engineering Distance Education program at an Indonesian polytechnic. The LED and LKPS documents are the post-audit version from the 2024 cycle. Single-program scope is intentional as a tractable initial study; multi-program generalization is acknowledged as a limitation.

Materials and Main Tools. The system is built with: (1) embedding model `paraphrase-multilingual-MiniLM-L12-v2` (384-dimensional, Indonesian-capable); (2) ChromaDB as the vector store with HNSW + cosine similarity over page-aware LED chunks; (3) LangGraph as the agent state-machine framework; (4) Ollama as the local inference backend on a 24 GB VRAM GPU server accessed via a Cloudflare-tunneled HTTPS endpoint; and (5) Claude (Anthropic) as the reference annotator. The end-to-end transformation from the manual workflow to the automated one is shown in Figure 1.

Figure 1 — Manual vs Automated Internal Quality Audit Workflow



Four Agent Tools. The agent has access to four LangChain tools: (a) `get_indicator_rubric(indicator_no)` returns the score criteria, formula, and LKPS table references; (b) `search_led_evidence(query, n_results)` performs vector search over LED chunks; (c) `get_lkps_table(table_reference)` returns the LKPS table as markdown; and (d) `compute_formula_score(formula_expression, variables)` is a sandboxed Python evaluator for formula-based indicators.

Two Agent Paradigms. Two paradigms are applied automatically based on the model-name pattern. The *native* paradigm (Qwen 3.5 and Mistral-Small) uses structured `tool_calls` directly executed by LangGraph's `ToolNode`. The *prompt-based* paradigm (DeepSeek-R1) instructs the model to emit single-line JSON of the form `{"action": ..., "args": ...}` for tool calls or `{"action": "final", "score": ..., "justification": ..., "confidence": ...}` for the final answer, with the `<think>...</think>` block stripped before JSON extraction.

Data Collection. Each indicator is evaluated once per model with identical sampling parameters (`temperature = 0.1`, `max_tokens = 8192`), producing 57 invocations (3 models × 19 indicators). Results are streamed to `benchmark_results.csv` per row for crash-durability.

Ground-Truth Construction (Validity Test). Because a regulator-issued gold standard for these 19 indicators is unavailable, the reference scores were generated through an LLM-as-annotator procedure using Claude-Sonnet (Anthropic, 2025), an external model unrelated to the three benchmarked open-weight LLMs. For each indicator the annotator received (a) the rubric text, (b) the relevant LED chunks pre-retrieved from ChromaDB, (c) the relevant LKPS sheets

parsed to text, and (d) a chain-of-thought prompt requiring an explicit evidence citation per scoring criterion. The annotator output was then spot-validated against the post-audit human assessor scores from the same 2024 cycle: 17 of 19 Claude scores (89.5%) agreed exactly with the human assessor, and 19 of 19 (100%) were within ±1, supporting use of the Claude annotations as the reference set for this study.

Statistical Analysis Techniques. The quantitative analysis uses six metrics: (1) **Mean Absolute Error (MAE):** $MAE = (1/n) \cdot \Sigma |y_i - \hat{y}_i|$; (2) **Root Mean Squared Error (RMSE):** $RMSE = \sqrt{(1/n) \cdot \Sigma (y_i - \hat{y}_i)^2}$; (3) **exact-match accuracy** as the fraction of indicators with $|y_i - \hat{y}_i| = 0$; (4) **within-±1 accuracy** as the fraction with $|y_i - \hat{y}_i| \leq 1$, the commonly accepted inter-assessor agreement threshold; (5) **Pearson r** as the linear correlation between model and reference scores; and (6) the **score-zero rate** as a proxy for the *false-refusal* failure mode. The analysis is performed both in aggregate and disaggregated by indicator type (qualitative / quantitative / composite) to surface per-type failure profiles.

RESULTS AND DISCUSSION

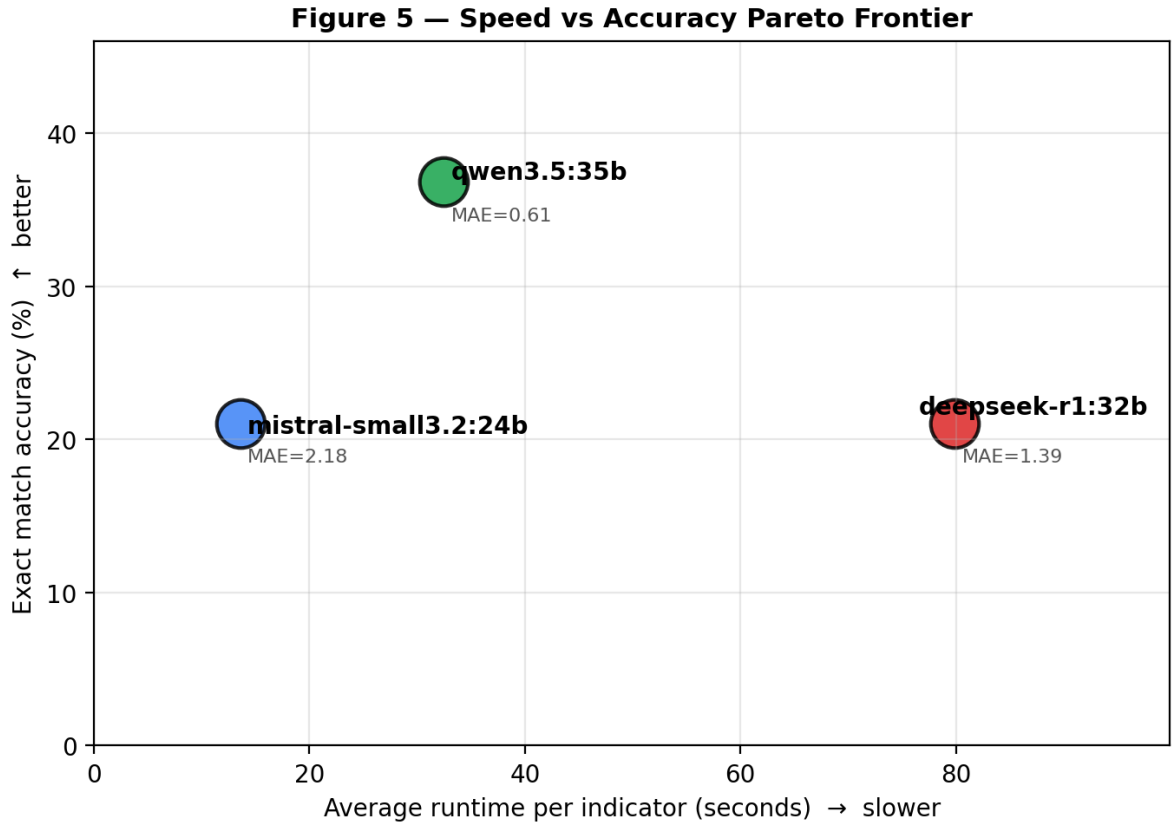
Aggregate Performance

Table 1 summarizes the aggregate performance of the three models on the 19-indicator sample. Two findings stand out. *First*, Qwen 3.5 dominates accuracy across every metric: lowest MAE (0.605), lowest RMSE (0.769), highest exact match (36.8%), and the only model with 100% within-±1 accuracy. *Second*, the runtime ordering inverts the prior 7-indicator finding: Mistral is now the fastest (13.6 s/indicator), DeepSeek-R1 the slowest (79.9 s/indicator) — the increased DeepSeek runtime is attributable to its extensive `<think>` deliberation on a broader indicator set.

Table 1. Aggregate Performance on 19 Indicators (n = 19 per model)

Model	MAE ↓	RMSE ↓	Exact ↑	Within-±1 ↑	Pearson r	Conf.	RT/ind
qwen3.5:35b	**0.605**	**0.769**	**36.8%**	**100.0%**	0.108	0.78	32.5 s
deepseek-r1:32b	1.395	1.825	21.1%	63.2%	0.148	0.83	79.9 s
mistral-small3.2:24b	2.184	2.598	21.1%	31.6%	0.204	0.53	13.6 s

DeepSeek-R1 has the highest average confidence (0.83) despite the second-largest error band — a clear *overconfidence* pattern previously documented for reasoning-distilled models (Geng et al., 2024). Mistral combines low average confidence (0.53) with the lowest within-±1 accuracy, indicating that even its low-confidence outputs are often destructive (score 0). Pearson correlations are uniformly weak (0.108–0.204), so LLM scores should be treated as candidate evidence for human review rather than as a stand-alone ranking. The speed-versus-accuracy trade-off is visualized in Figure 2.



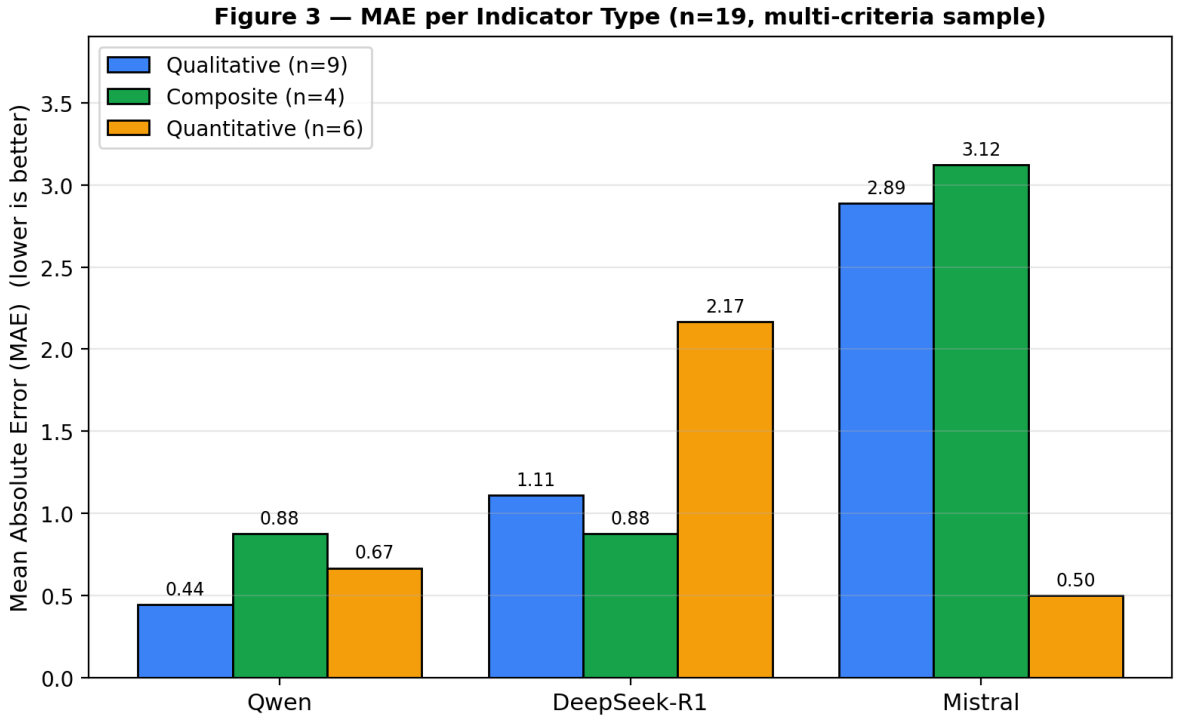
Per-Type Breakdown

Table 2 and Figure 3 disaggregate the performance by indicator type. The three models reveal sharply different per-type failure profiles.

Table 2. Per-Type Breakdown of Model Performance

Model	Type	n	MAE ↓	Exact ↑	±1 ↑	Score=0
qwen3.5:35b	Qualitative	9	**0.44**	**55.6%**	**100.0%**	0
qwen3.5:35b	Quantitative	6	0.67	33.3%	**100.0%**	0
qwen3.5:35b	Composite	4	0.88	0.0%	**100.0%**	0
deepseek-r1:32b	Qualitative	9	1.11	33.3%	77.8%	1
deepseek-r1:32b	Quantitative	6	2.17	0.0%	33.3%	2
deepseek-r1:32b	Composite	4	0.88	25.0%	75.0%	0
mistral-small3.2:24b	Qualitative	9	2.89	0.0%	11.1%	**7**
mistral-small3.2:24b	Quantitative	6	**0.50**	**66.7%**	83.3%	0
mistral-small3.2:24b	Composite	4	3.12	0.0%	0.0%	3

Qwen 3.5 maintains 100% within-±1 across all three types — it is the only model with a balanced per-type profile. DeepSeek-R1 behaves *opposite* to the conventional expectation that reasoning models excel at numbers: its quantitative MAE (2.17) is twice its qualitative MAE (1.11), driven by failures on `RBK` (indicator 30) and `RIPK` (indicator 42), where the model retrieved the correct LKPS rows but its `<think>` block reasoned itself into a "data is insufficient" refusal. Mistral shows the most dramatic inversion: 7 of 9 qualitative indicators receive score 0 (false-refusal rate ≈ 78%) while its quantitative MAE is the lowest of all three models at 0.50. Inspection of justifications shows that Mistral often returns instructional output ("To score this indicator, an evaluator should…") rather than an actual score on qualitative prompts. The per-type MAE pattern is visualized in Figure 3.



Failure Mode: False Refusal

Across the 19 indicators, the dominant failure mode is *false refusal* — the model emits a score of 0 with the justification "no evidence" even when LED chunks containing the evidence are retrieved. Table 3 summarizes the false-refusal rate per model.

Table 3. False-Refusal Rate per Model

Model	Total score = 0	Affected indicators
qwen3.5:35b	**0**	(none)
deepseek-r1:32b	3	30, 42, 48
mistral-small3.2:24b	**10**	1, 2, 4, 7, 8, 13, 17, 22, 39, 48, 53, 60 (qual + composite)

Qwen 3.5 shows perfect reliability — it never refuses to score. DeepSeek-R1 has three false refusals, with chain-of-thought inspection revealing over-cautious phrasing such as "evidence is too generic to verify all four aspects". Mistral exhibits severe false refusal on 10 of 19 indicators, consistent with the *over-cautious instruction-tuning* pattern previously reported on Indonesian-language tasks (Nurohim et al., 2025).

Discussion: Comparison with Prior Work

Compared to Aljarallah and Dutta (2022), who built a rule-based "Quality Automation Framework" for Saudi accreditation specifications, the present agent removes the maintenance burden of rule encoding by delegating evidence retrieval to a learned retriever and scoring to an LLM. Compared to Sakinah et al. (2025), who used K-Means + OLAP for Indonesian accreditation analytics, our system operates on the source-of-truth documents (LED + LKPS) rather than on aggregated metrics, eliminating an ETL hop that loses fine-grained evidence. Compared to Demir and Yavuz (2026), a generic comparative LLM study on multidimensional data, this work narrows scope to a regulated audit task with verifiable score-rubric grounding, where the generative-model strengths they report translate into both opportunities (Qwen) and risks (Mistral hallucination). Compared to the AI-assisted self-evaluation frameworks proposed at a conceptual level by Başaran (2026) and Ticolau (2026), this study contributes an end-to-end implementation with a reproducible benchmark and an open per-indicator error log. The trade-off identified here — Qwen's reliability against DeepSeek's overconfidence and Mistral's qualitative collapse — is consistent with the calibration literature on reasoning-tuned and instruction-tuned mid-sized LLMs (Lin et al., 2022; Geng et al., 2024) but extends it to Indonesian accreditation documents, a domain not previously covered.

Implication for Practice

For deployment as an AI draft scorer to be verified by a human assessor, Qwen 3.5 35B is recommended because its 100% within-±1 accuracy means the assessor only needs to correct at most one scale point. DeepSeek-R1 may be used for composite-heavy batches with awareness of occasional false refusals on quantitative items. Mistral-Small 3.2 should be avoided for qualitative-heavy batches because its 78% false-refusal rate would require extensive manual correction; it remains useful for quantitative-only fast scoring (13.6 s per indicator, MAE 0.50). This recommendation aligns with the *human-in-the-loop assistive tool* framing: the final decision remains with the certified assessor, and the LLM agent serves only as an accelerator that produces verifiable draft scores and justifications.

Limitations

Four limitations bound this study. First, the reference set is itself LLM-generated (Claude-Sonnet) and validated on a single post-audit program; future work should add multi-program human gold scores. Second, n = 19 is purposively stratified for diagnostic per-type analysis, so the reported MAE/RMSE figures are descriptive of behavior on a stratified diagnostic set rather than a population point estimate. Third, the benchmark uses a single deterministic-leaning sampling configuration;

sensitivity to `temperature` and `top_p` is not explored. Fourth, the study is single-site, so cross-institution generalization is unverified.

CONCLUSION

This study developed and benchmarked a cross-modal RAG agent for Internal Quality Audit in Indonesian LAM-Teknik accreditation, evaluating three open-weight LLMs (Qwen 3.5 35B, DeepSeek-R1 32B, Mistral-Small 3.2 24B) on 19 indicators sampled across all nine criteria. Qwen 3.5 is the only model that satisfies operational accuracy across qualitative, quantitative, and composite types simultaneously (MAE 0.605, 100% within ±1, 32.5 s per indicator). DeepSeek-R1 is the most cautious in qualitative items but is the slowest and collapses on quantitative indicators (MAE 2.17). Mistral-Small is the fastest (13.6 s) and surprisingly strong on quantitative items (MAE 0.50) yet fails 78% of qualitative items by emitting a score-zero refusal. End-to-end, the agent reduces AMI cycle time from 3–5 days to under one hour per program while preserving a fully auditable evidence trace for human verification. For practical deployment as an AI draft scorer the study recommends Qwen 3.5 35B because of its consistent reliability. Future work will (1) extend the benchmark to the full 60-indicator instrument across multiple study programs; (2) replace the Claude-based reference with multi-program human gold scores; and (3) introduce a refusal-suppression filter and a confidence-recalibration head for production deployment.

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